

Will Maberry

 will-maberry |  willmaberry.com |  will-maberry

RESEARCH INTERESTS

Computational epidemiology; spatiotemporal modeling; infectious disease forecasting; rare-event detection; One Health; biosurveillance; geospatial ML

EDUCATION

Mississippi State University

M.S. Vet. & Biomed. Science Aug. 2026 – May 2028
Computational Biology Concentration
Fully funded research appointment via USDA ARS
Research Advisor: Dr. Isaac Jumper

The University of Texas at Arlington (UTA)
B.S. Computer Science Aug. 2022 – May 2026

HONORS & CERTIFICATIONS

- Summa Cum Laude: UTA (GPA 3.824)
- Judge's Choice Award: UTA College of Engineering Innovation Day (2026), for 2025 internship research
- Fundamentals of AI Certificate: UTA

TECHNICAL SKILLS

Forecasting & Modeling

Spatiotemporal forecasting; automated leakage-safe ML pipelines; rare-event modeling; ablation studies

Machine Learning

PyTorch; TensorFlow; Scikit-learn; Imbalanced-learn; Optuna; AutoGluon

Geospatial & Scientific Computing

GeoPandas; Google Earth Engine; raster processing

Programming & Data Engineering

Python; C; Java; SQL; Bash; Pandas; NumPy

Pipelines & Deployment

GitHub/Git, FastAPI; Flask; Docker

Visualization

Matplotlib; Folium/Leaflet; interactive dashboards

RELEVANT COURSEWORK

Machine Learning & AI

Machine Learning (ML); Neural Networks & Deep Learning; Artificial Intelligence (AI); Computer Vision

Data & Algorithms

Data Analysis & Modeling Techniques; Design & Analysis of Algorithms; Algorithms & Data Structures

Mathematics

Calculus I-III; Linear Algebra; Discrete Structures; Theoretical Computer Science

Graduate Coursework (In Progress)

Applied Veterinary Epidemiology; Management of Commercial Layers; Statistical Methods

EXPERIENCE

USDA ARS — Research Assistant Jul. 2025 – Jul. 2026
Part-Time Employee

- Built a **leakage-safe, forward-looking** county-month HPAI forecasting framework for the continental U.S., enforcing forward-only temporal truncation and forecast-month covariate synthesis.
- Evaluated **40 model configurations** (graph-based spatiotemporal models and gradient boosters) via county adjacency networks across a 24-month rolling-origin backtest spanning **74,616 county-month forecasts**.
- Achieved **73% balanced accuracy and 74% recall** against 0.8% outbreak-positive prevalence, correctly identifying 417 of 563 outbreak-positive county-months.

- Deployed forecasts into a decision-support web application translating county-level risk into interpretable biosurveillance tiers for producers and animal-health professionals.

USDA ARS — Research Intern May 2025 – Jul. 2025
Summer Internship (2025)

- Developed national-scale county-level HPAI risk classification and pseudo-forecasting models using environmental and climate covariates across the continental U.S.
- Achieved **75%+ balanced accuracy** on highly imbalanced (~1% positive) outbreak data spanning 2022–2024.
- Presented research findings to the **USDA ARS Chief Scientist** and multiple National Program Leaders (NPLs) to support disease biosurveillance.

OpenAI — Engagement Manager Jun. 2024 – Jun. 2026
Contract Employee

- Scaled OpenAI's official Discord community to **850k+ users** (680% growth) through technical programming, online events, and community engagement initiatives.

TEACHING & COMMUNITY

- Teaching Assistant (Operating Systems):** Supported 120+ students through office hours and review sessions.
- OpenAI Community Volunteer:** Supported global AI education initiatives and technical discussions (2022–2024), later transitioning to a paid contract role.
- Association for Computing Machinery (ACM):** Education Director for 1500+ member organization; led 15+ professional and educational workshops.
- The Wesley Foundation @ UTA:** Sole student representative on Board of Directors overseeing \$100k+/year budget.

PUBLICATIONS

Blackstock, J., Maberry, W., et al.

“A forward-looking high pathogenicity avian influenza monthly risk model for counties in the conterminous United States”
Manuscript submitted to Scientific Reports (May 2026)